Tô Gia Bảo

Data Engineer | AML Analyst

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DETAILED WORK EXPERIENCE

Data Engineering | AML & Fraud Analytics

Timo Digital Bank by BVBank — Ho Chi Minh City, Vietnam
Data Engineer — Mar. 2025 – Present

Role Overview

Build data pipelines, analytical models, and operational reporting for digital banking. Responsibilities span ingestion, PII masking, data quality, and storage optimization, alongside transaction monitoring and risk analytics to support Compliance and Fraud Operations.

01 · Data Engineering

MariaDB Data Pipelines

Implemented a workflow from MariaDB ingestion to curated datasets and downstream analytical transformations.

- Built ingestion workflows using Amazon EMR and AWS Glue to bring source data into the processing pipeline.
- Applied PII masking in the curated layer to protect personally identifiable information before downstream use.
- Developed dbt transformations to turn processed data into analytics-ready datasets for reporting and analysis.

Deliverable: An integrated ingestion, masking, and transformation workflow supporting data consumption.

S3 Storage Optimization

Optimized a large S3 table using Parquet ZSTD-3 compression and file compaction to reduce storage volume and object count.

- Applied compression and consolidated files across 56.8 billion rows covering 2023 to September 2026.
- Reduced storage by **38.4 TB (45%)** and object count by **80%** for the optimized table.

Data Modeling and Operational Reporting

- Built dbt fact and dimension models with defined grains, keys, and relationships.
- Powered Lightdash dashboards for Airflow reliability, SLA tracking, Athena usage, and S3 storage health.

Deliverable: Analytical models and dashboards for monitoring data platform operations.

Data Quality and Integration Testing

- Used dbt tests, Great Expectations, and pytest to validate data transformations and integrations.
- Implemented regression checks for query contracts, SQL compilation, and dbt/Lightdash metadata consistency.

02 · AML and Fraud Analytics

Behavioral Feature Mart

Built a daily incremental feature mart on Amazon Athena and dbt to support transaction monitoring and account risk assessment.

- Developed **40+ behavioral features** across millions of transactions.
- Captured indicators of fund pass-through, structuring, and unusual transaction arrival patterns for downstream risk analysis.

Mule Account Risk Scoring

Developed a **10-signal risk scoring engine** to identify potential mule accounts and prioritize review.

- Combined indicators of transaction velocity, nighttime bursts, small incoming transfers, and concentrated cash-outs.
- Added balance retention gates and two-tier review queues to improve alert prioritization.

- Reduced false-positive triage workload for Compliance and Fraud Operations by **approximately 50%**, while retaining confirmed mule cases for review.

Transaction Anomaly Alerts

- Compared intraday and month-to-date transaction frequency and gross transaction value against a three-month historical baseline.
- Built daily alerts to flag sudden behavioral spikes for further review.

Deliverables: Behavioral features, risk scores, prioritized review queues, and daily alerts supporting Compliance and Fraud Operations.

03 · Analytics Tools and Operational Automation

Self-Service Analytics in Slack

Built a FastMCP server for conversational analytics in Slack, using dbt documentation, Lightdash dashboards, and semantic metadata to support self-service data discovery.

Workflow Automation and Observability

- Automated MongoDB backup validation and general ledger lookups with Python and Airflow, reducing manual checks.
- Migrated Timo Metrics from Loki/Grafana to SigNoz, centralizing logs and metrics for operational monitoring and troubleshooting.
- Delivered Tracking247 using Streamlit and Airflow to accept shipment IDs from the Customer team and synchronize shipment status into MongoDB.